

(A bit sloppy this time)

$$3.4 \#6 \quad \begin{cases} u_{tt} - c^2 u_{xx} = f(x, t) & (1) \\ u(x, 0) = \phi(x) \\ u_t(x, 0) = \psi(x) \end{cases}$$

(a) Rewrite (1) as $(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x}) \underbrace{(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x}) u}_{\stackrel{\text{def}}{=} v(x, t)} = f(x, t)$.

$$\Rightarrow \begin{cases} u_t + c u_x = v \\ v_t - c v_x = f. \end{cases}$$

(b) Method of characteristics.

$$\begin{cases} \frac{dt}{d\tau} = 1 \\ \frac{dx}{d\tau} = c \\ \frac{dv}{d\tau} = v \end{cases} \quad \text{s.t.} \quad \begin{cases} t(0) = 0 \\ x(0) = C \\ v(0) = f(C). \end{cases}$$

$$\text{Thus } \begin{matrix} t = \tau + C_1 & \xrightarrow{t(0)=0} & t = \tau \\ x = c\tau + C_2 & \xrightarrow{x(0)=C} & x = c\tau + C \end{matrix} \quad \} \Rightarrow C = x - ct.$$

$$\frac{dv}{d\tau} = v(x, t) = v(c\tau + C, \tau)$$

$$\Rightarrow v(\tau) = \int_0^\tau v(cs + C, s) ds + C_3$$

Since $v(0) = f(C)$, $C_3 = f(C)$.

$$\text{Thus } v(\tau) = \int_0^\tau v(cs + C, s) ds + f(C)$$

Describing τ, C by x, t , i.e. $\tau = t$, $C = x - ct$, then

$$u(x, t) = \int_0^t v(cs + x - ct, s) ds + f(x - ct).$$

$$= \int_0^t v(x-ct+cs, s) ds \quad \text{if we choose } f=0.$$

(c) Similarly, $v(x, t) = \int_0^t f(x+ct-cz, z) dz$

(d) $u(x, t) = \int_0^t v(x-ct+cs, s) ds$

$$= \int_0^t \int_0^s f(x-ct+cs+cs-cz, z) dz ds$$

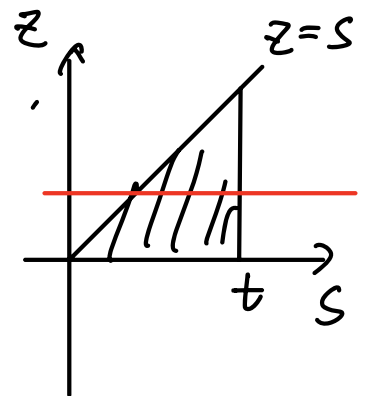
$$\text{Fubini} = \int_0^t \int_z^t f(x-ct+cs+cs-cz, z) ds dz$$

$$y = x-ct+2cs-cz$$

$$dy = 2c ds$$

$$= \int_0^t \int_{x-ct+cz}^{x+ct-cz} f(y, z) \frac{1}{2c} dy dz$$

$$= \frac{1}{2c} \int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y, s) dy ds$$



NB: I describe the shaded region $\{0 \leq z \leq s\} \cap \{0 \leq s \leq t\}$ in two ways, using Fubini's theorem.

4.2 #4 Consider $\begin{cases} u_t - k u_{xx} = 0 & (1) \quad -l \leq x \leq l \\ u(-l, t) = u(l, t) & (2) \\ u_x(-l, t) = u_x(l, t) & (3) \end{cases}$

(a) Show that the eigenvalues are $\lambda = \left(\frac{n\pi}{2l}\right)^2, n=0, 1, \dots$

Proof. Say $u(x, t) = X(x)T(t)$, then

by (1), $X T_t - k X_{xx} T = 0 \Rightarrow \frac{X_{xx}}{X} = \frac{T_t}{kT} \stackrel{\text{def}}{=} -\lambda$

By (2), $X(-l)T(t) = X(l)T(t) \Rightarrow X(-l) = X(l)$

By (3), $X_x(-l)T(t) = X_x(l)T(t) \Rightarrow X_x(-l) = X_x(l)$

(i) $-\lambda = \mu^2 > 0$

$$\cosh(x) := \frac{e^x + e^{-x}}{2}$$

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$X = A \cosh(\mu x) + B \sinh(\mu x)$$

$$X(l) = X(-l) \Rightarrow A \cosh(\mu l) + B \sinh(\mu l) = A \cosh(\mu l) + B \sinh(\mu l)$$

$$\Rightarrow B \sinh(\mu l) = 0 \Rightarrow B = 0$$

$$X_x(l) = X_x(-l) \Rightarrow A \mu \sinh(\mu l) + B \mu \cosh(\mu l) = A \mu \sinh(\mu l) + B \mu \cosh(\mu l)$$

$$\Rightarrow A \mu \sinh(\mu l) = 0 \Rightarrow A = 0 \text{ since } \mu > 0 \text{ and } \mu l > 0$$

$$X \equiv 0 \Rightarrow -\lambda \text{ cannot be positive.}$$

(ii) $-\lambda = 0, X_{xx} = 0 \Rightarrow X = Ax + B$

$$X(l) = X(-l) \Rightarrow Al + B = A(-l) + B \Rightarrow A = 0$$

$$X_x(l) = X_x(-l) \Rightarrow A = A, \text{ which is redundant}$$

Thus $X \equiv B$ is a constant, and $\frac{T'}{kT} = 0$ implies $T \equiv C$ is also a constant.

$$(iii) -\lambda = -\mu^2 < 0.$$

$$X = A \cos(\mu x) + B \sin(\mu x)$$

$$X(l) = X(-l) \Rightarrow A \cancel{\cos(\mu l)} + B \sin(\mu l) = A \cancel{\cos(-\mu l)} + B \sin(-\mu l) \\ \Rightarrow B \sin(\mu l) = 0$$

$$X_x(l) = X_x(-l) \Rightarrow \mu A \sin(\mu l) + \mu B \cancel{\cos(\mu l)} = -\mu A \sin(-\mu l) + \mu B \cancel{\cos(-\mu l)} \\ \Rightarrow \mu A \sin(\mu l) = 0 \Rightarrow \text{since } \mu \neq 0, A \sin(\mu l) = 0.$$

$$\text{Thus } \sin(\mu l) = 0 \Rightarrow \mu l = n\pi \quad n \in \mathbb{N} \Rightarrow \mu_n = \frac{n\pi}{l} \\ \Rightarrow \lambda_n = (\mu_n)^2 = \left(\frac{n\pi}{l}\right)^2$$

$$\frac{T_t}{kT} = -\lambda = -\left(\frac{n\pi}{l}\right)^2 \Rightarrow T_t + k\left(\frac{n\pi}{l}\right)^2 T = 0 \\ \Rightarrow T = D \cdot e^{-k\left(\frac{n\pi}{l}\right)^2 t} \text{ for some } D$$

The eigenvalues are $\lambda = 0, \left(\frac{n\pi}{l}\right)^2$
 given by (ii) given by (iii)

(b) Show that

$$u(x, t) = \frac{1}{2} A_0 + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi x}{l} + B_n \sin \frac{n\pi x}{l} \right) e^{-n^2 \pi^2 k t / l^2}.$$

$$\text{Proof. } u(x, t) = \sum_{n=0}^{\infty} X_n(x) T_n(t) \\ = X_0(x) T_0(t) + \sum_{n=1}^{\infty} X_n(x) T_n(t) \\ = B \cdot C + \sum_{n=1}^{\infty} \left(A'_n \cos \left(\frac{n\pi}{l} x \right) + B'_n \sin \left(\frac{n\pi}{l} x \right) \right) \cdot D_n \cdot e^{-k \left(\frac{n\pi}{l} \right)^2 t}$$

$$= \frac{1}{2} A_0 + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi x}{l} + B_n \sin \frac{n\pi x}{l} \right) e^{-n^2 \pi^2 k t / l^2}$$

with a proper choice of constants, i.e., $\frac{1}{2} A_0 \stackrel{\text{def}}{=} B \cdot C$, $A_n \stackrel{\text{def}}{=} A'_n \cdot D_n$, $B_n \stackrel{\text{def}}{=} B'_n \cdot D_n$

4.3 #4 Consider the Robin problem

$$\begin{cases} -X'' = \lambda X \\ X'(0) = a_0 X(0) \\ X'(l) = -a_l X(l) \end{cases}$$

If $a_0 < 0$, $a_l < 0$ and $-a_0 a_l < a_0 a_l l$,

show that there are two negative eigenvalues.

(i) $-\lambda = \mu^2 > 0$

$$X = A \cosh(\mu x) + B \sinh(\mu x)$$

$$X'(0) = a_0 X(0)$$

$$\Rightarrow A \mu \sinh(\mu \cdot 0) + B \mu \cosh(\mu \cdot 0) = a_0 (A \cosh(\mu \cdot 0) + B \sinh(\mu \cdot 0))$$

$$\Rightarrow B \mu = a_0 A$$

$$X'(l) = -a_l X(l)$$

$$\Rightarrow A \mu \sinh(\mu \cdot l) + B \mu \cosh(\mu \cdot l) = -a_l (A \cosh(\mu \cdot l) + B \sinh(\mu \cdot l))$$

$$\Rightarrow (A \mu + a_l B) \cdot \sinh(\mu l) = -(a_l A + B \mu) \cosh(\mu l)$$

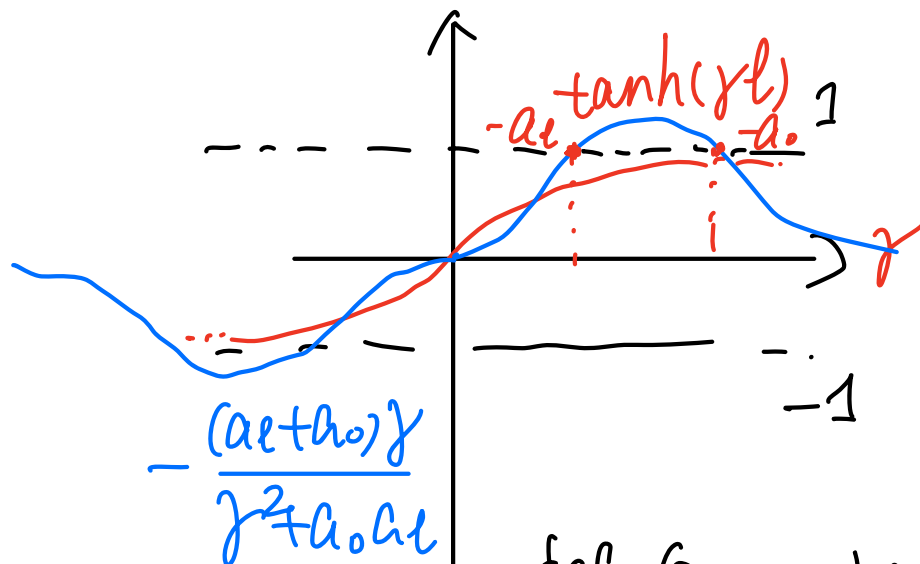
$$\Rightarrow \tanh(\mu l) = - \frac{a_l A + B \mu}{A \mu + a_l B} \stackrel{B \mu = a_0 A}{=} - \frac{a_l A + a_0 A}{A \mu + a_l \cdot \frac{a_0 A}{\mu}}$$

$$= - \frac{\mu A (a_l + a_0)}{A \mu^2 + a_0 a_l A} = - \frac{\mu (a_l + a_0)}{\mu^2 + a_0 a_l}$$

(if $A = 0$, then by $B \mu = a_0 A$ and $\mu > 0$, we have $B = 0$ and hence $X \equiv 0$, contradiction.)

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$



By the hint, if $f(\gamma) \stackrel{\text{def}}{=} -\frac{(a_0 + a_l) \cdot \gamma}{\gamma^2 + a_0 a_l} = 1$,

then $\gamma^2 + (a_0 + a_l)\gamma + a_0 a_l = 0$

$$\begin{aligned} \gamma_{1,2} &= \frac{-(a_0 + a_l) \pm \sqrt{(a_0 + a_l)^2 - 4a_0 a_l}}{2} \\ &= \frac{-(a_0 + a_l) \pm |a_0 - a_l|}{2} \end{aligned}$$

We may assume without loss of generality that $a_0 \leq a_l < 0$, then $|a_0 - a_l| = a_l - a_0$,

and then $\gamma_1 = -a_0$, $\gamma_2 = -a_l$

$\Rightarrow f(\gamma)$ crosses $y=1$ in two places

Moreover, $\frac{df}{d\gamma} = -(a_0 + a_l) \cdot \frac{\gamma^2 + a_0 a_l - \gamma \cdot 2\gamma}{(\gamma^2 + a_0 a_l)^2}$

$$= -(a_0 + a_l) \cdot \frac{a_0 a_l - \gamma^2}{(\gamma^2 + a_0 a_l)^2}$$

$$\frac{df}{d\gamma} = 0 \Leftrightarrow a_0 a_l - \gamma^2 = 0 \Leftrightarrow \gamma = \pm \sqrt{a_0 a_l}.$$

$\Rightarrow f(\gamma)$ has a single maximum.

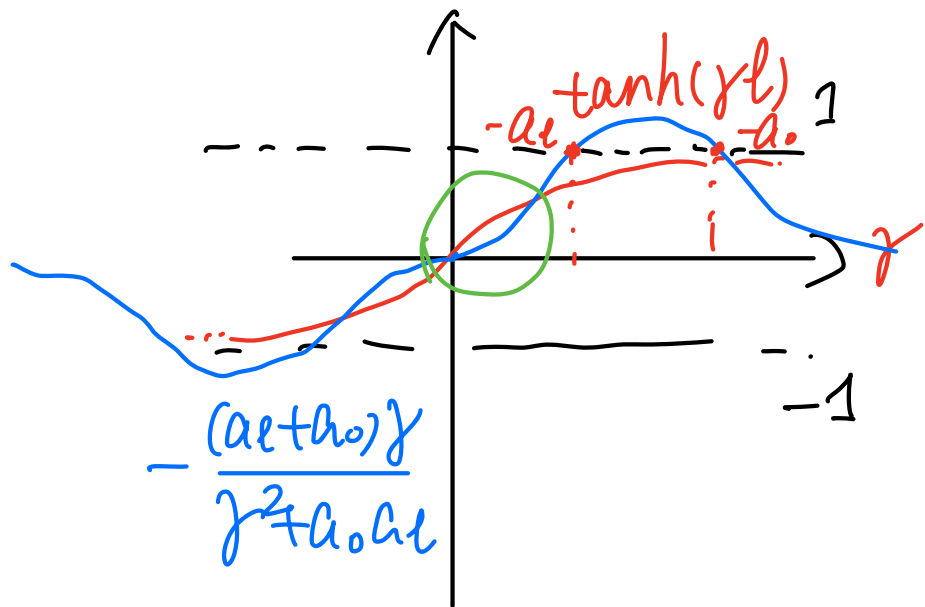
(You may just state $f''(\sqrt{a_0 a_1}) < 0$

and $f''(-\sqrt{a_0 a_1}) > 0$ without proof)

Finally, since $(\tanh(\gamma \cdot l))'|_{\gamma=0} = l$

$$\text{and } f'(\gamma)|_{\gamma=0} = -\frac{a_0 + a_1}{a_0 \cdot a_1}.$$

By the given condition $-a_0 - a_1 < a_0 a_1 \cdot l$,
we conclude the **red line** is above the **blue line**
when $\gamma > 0$ is small, i.e. the $\odot$ holds



We are done.

4.3 #6 Consider the Robin problem

$$\begin{cases} -X'' = \lambda X \\ X'(0) = a_0 X(0) \\ X'(l) = -a_l X(l) \end{cases}$$

If $a_0 = a_l = a$, show that:

$$(1) \text{ \# of negative eigenvalues} = \begin{cases} 0 & \text{if } a \geq 0 \\ 1 & \text{if } -\frac{2}{l} < a < 0 \\ 2 & \text{if } a < -\frac{2}{l} \end{cases}$$

By the same discussion as before, if $-\lambda = \mu^2 > 0$, then

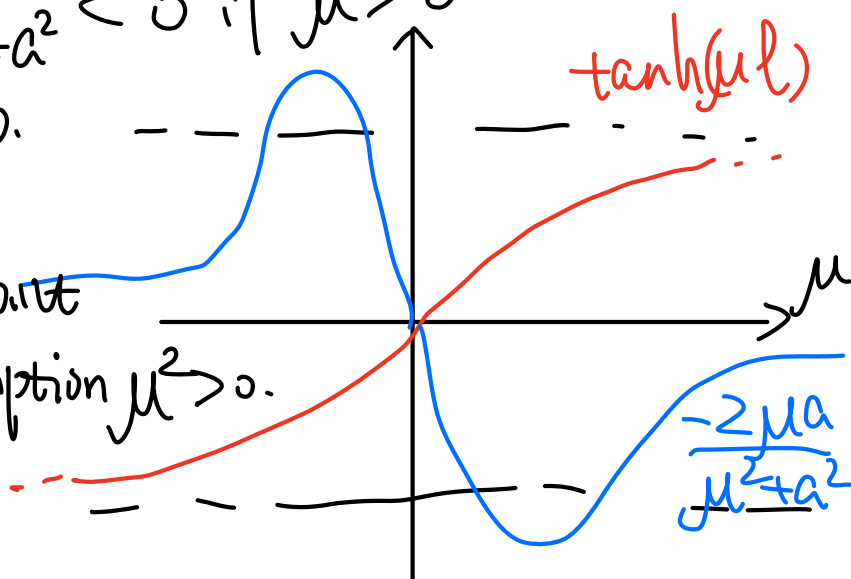
$$\tanh(\mu l) = -\frac{\mu(a_l + a_0)}{\mu^2 + a_0 \cdot a_l} = -\frac{2\mu a}{\mu^2 + a^2}.$$

(1') If $a \geq 0$, then $-\frac{2\mu a}{\mu^2 + a^2} < 0$ if $\mu > 0$

$$-\frac{2\mu a}{\mu^2 + a^2} > 0 \text{ if } \mu < 0.$$

The only intersection point is $\mu = 0$. But by assumption $\mu^2 > 0$.

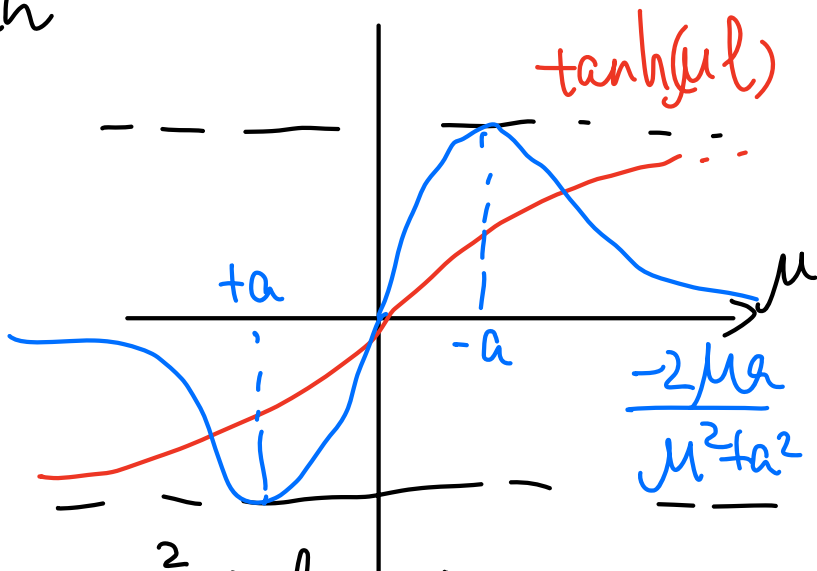
Contradiction



(2') If $-\frac{2}{l} < a < 0$, then we may verify

$$\frac{d}{d\mu} \left(\frac{-2\mu a}{\mu^2 + a^2} \right) \Big|_{\mu=0} = -\frac{2}{a}.$$

$$\frac{d}{d\mu} (\tanh(\mu l)) \Big|_{\mu=0} = l.$$



Since $-\frac{2}{l} < a < 0$, we have $-\frac{2}{a} > l$, i.e., the blue curve is above the red curve near the origin for $x > 0$. Also one can draw the graph of $\frac{-2\mu a}{\mu^2 + a^2}$ which is increasing over $0 < x < -a$ is decreasing over $x > -a$.

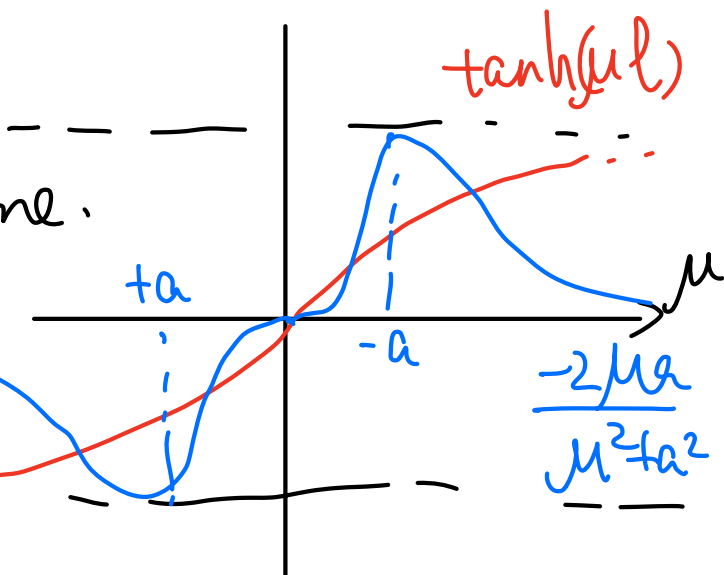
$\Rightarrow \tanh(\mu l) = \frac{-2\mu a}{\mu^2 + a^2}$ has only one positive root.

$\Rightarrow$ # of neg eigenvalue = 1 if $-\frac{2}{l} < a < 0$

(3') If $a < -\frac{2}{l}$, then similar as (2'),

we can draw the corresponding graph. But this time we have two positive roots.

Since $-\frac{2}{a} > l$ this time.



$\Rightarrow$ # of neg eigenvalue = 2 if $a < -2/l$.

(2) 0 is an eigenvalue iff $a=0$ or $a=-\frac{2}{l}$.

If 0 is an eigenvalue then $X''=0$

$$\Rightarrow X(x) = Ax + B, \quad A, B \text{ TBD}$$

$$\text{By } X'(0) = a_0 X(0), \quad A = a_0 B = aB$$

$$\text{By } X'(l) = -a_l X(l) \quad A = -a_l(Al + B) = -a(Al + B)$$

If $a=0$, we are done

If $a \neq 0$, then plugging $B = A/a$ into $A = -a(Al + B)$

$$\text{gives } A = -a(Al + \frac{A}{a}) \Rightarrow a = -\frac{2}{l}$$

(if $A=0$, then $B=0$, and $X(x) \equiv 0$, contradiction)

$$\text{Conversely, if } a=0, \text{ then } \begin{cases} -X'' = \lambda X \\ X'(0) = X'(l) = 0 \end{cases}$$

$\Rightarrow X \equiv C$ is an eigenvector corresponding to the eigenvalue $\lambda=0$

$$\text{If } a = -\frac{2}{l}, \text{ then } \begin{cases} -X'' = \lambda X \\ X'(0) = -\frac{2}{l} X(0) \\ X'(l) = \frac{2}{l} X(l) \end{cases}$$

$\Rightarrow X(x) = -\frac{2}{l}x + 1$ is an eigenvector corresponding to the eigenvalue $\lambda=0$.

4.3 #7 Consider the Robin problem

$$\begin{cases} -X'' = \lambda X \\ X'(0) = a_0 X(0) \\ X'(l) = -a_l X(l) \end{cases}$$

If $a_0 = a_l = a$, show that:

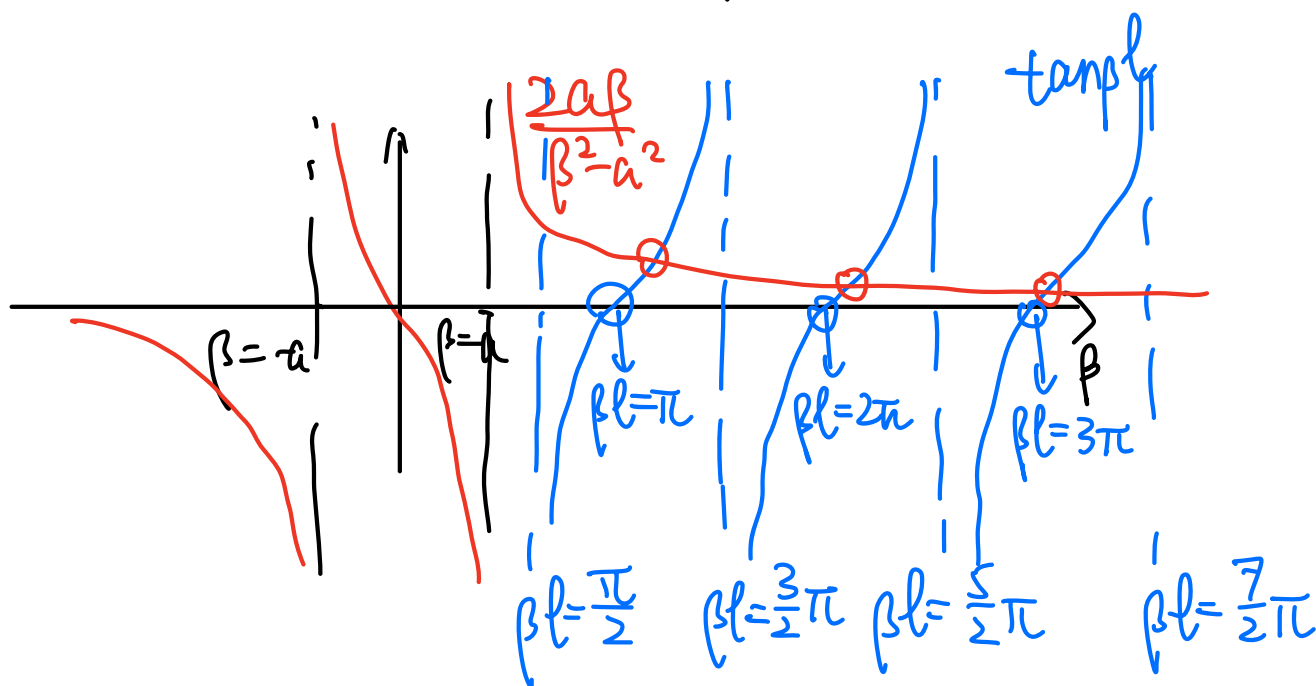
$$\lim_{a \rightarrow +\infty} (\beta_n(a) - \frac{(n+1)\pi}{l}) = 0,$$

where $\lambda_n(a) = [\beta_n(a)]^2$ is the $(n+1)$ -st eigenvalue.

Proof: By the discussion in Strauss, Page 94, case 1

if $\lambda = \beta^2 > 0$, then β satisfies the equation

$$\tan(\beta l) = \frac{(a_0 + a_l)\beta}{\beta^2 - a_0 a_l} = \frac{2a\beta}{\beta^2 - a^2} \quad (*)$$



It's equivalent to showing $\lim_{a \rightarrow +\infty} \beta_n(a) \cdot l - (n+1)\pi = 0$

It simply means $\circ^0$ are closer and closer as $a \rightarrow +\infty$, for a fixed n . (Warning: we are not saying $\beta \rightarrow \infty$!)

Fix an integer $n \geq 0$.

We compute $\beta_n(a)l - (n+1)\pi$

$$\text{by } (*) = \left(\arctan\left(\frac{2a\beta_n}{\beta_n^2 - a^2}\right) + (n+1)\pi \right) - (n+1)\pi$$

$$= \arctan\left(\frac{2a\beta_n}{\beta_n^2 - a^2}\right)$$

$$\lim_{a \rightarrow \infty} \arctan\left(\frac{2a\beta_n}{\beta_n^2 - a^2}\right) = \arctan\left(\lim_{a \rightarrow \infty} \frac{2a\beta_n}{\beta_n^2 - a^2}\right)$$

$$= \arctan(0) = 0.$$

NB: You may worry $\beta_n = \beta_n(a)$ is also a function in a , but it would be fine, since

$$\beta_n(a) \sim \frac{n\pi}{l},$$

$$\text{i.e., } C_2(n) \leq \beta_n(a) \leq C_1(n)$$

such that C_1, C_2 only depend on n , so that

$$0 \leq \left| \frac{2a\beta_n}{\beta_n^2 - a^2} \right| \leq \left| \frac{2aC_1(n)}{C_1(n)^2 - a^2} \right| \rightarrow 0 \text{ as } a \rightarrow +\infty.$$

(Note we are fixing an n .)